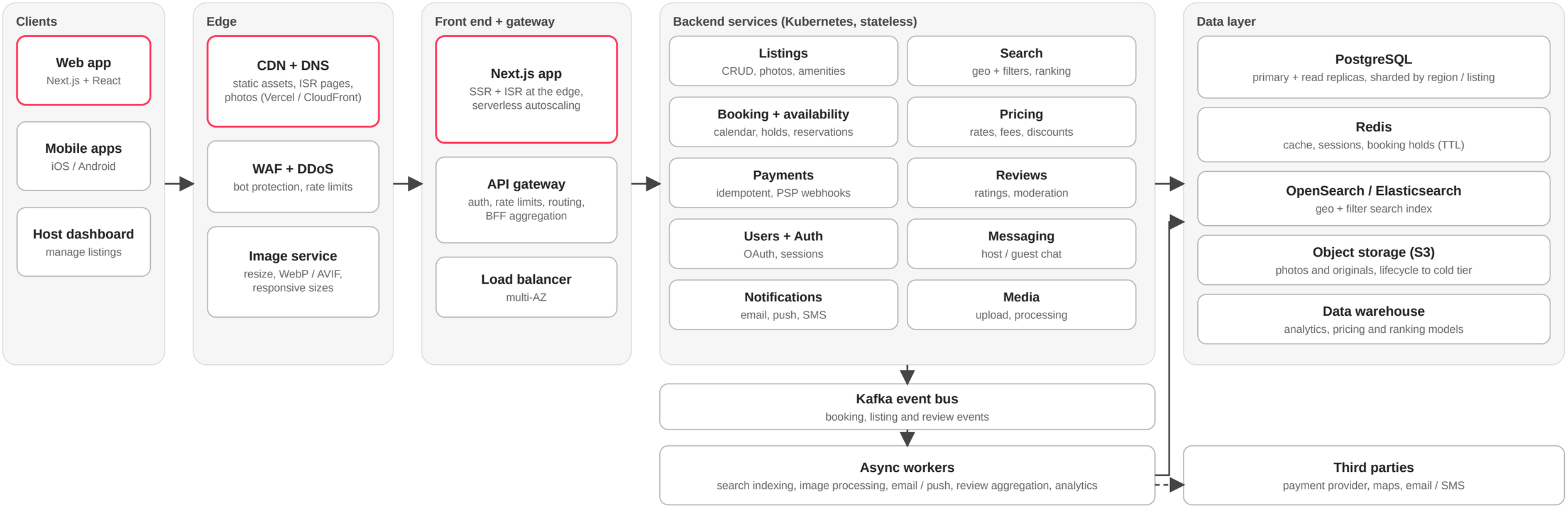
Production architecture: vacation-rental marketplace (Airbnb-style)

Request path left to right. Pink outline = the part built in this take-home (Next.js front end on Vercel with local data).



Scaling strategy by layer

Front end

- Static pages + ISR cached on the CDN
- Images: responsive sizes, WebP / AVIF, lazy load
- Code-split routes; render at the edge near users
- Skeletons and optimistic UI for Save / Reserve

Backend

- Stateless services, horizontal autoscaling (HPA)
- One service per domain, scaled independently
- Queues absorb booking and email spikes
- Idempotent APIs, retries, circuit breakers

Storage

- Postgres primary + read replicas, shard by region
- Redis for hot listings and availability
- Booking: transactions + short holds, no double-booking
- S3 + CDN for photos; archive old media

Search

- OpenSearch with geo and filter indexes
- Sharded and replicated across zones
- Fed from Kafka so writes never block search
- Cache popular queries; separate ranking service

Deployment

- CI/CD: tests, build, container, canary / blue-green
- Multi-AZ, multi-region failover; Terraform for infra
- Metrics, logs, traces and SLO alerts
- Secrets manager; least-privilege access

Booking flow: gateway, Booking service takes a Redis hold, Payments charges through the provider, Postgres commits the reservation, an event on Kafka triggers search re-index and notifications.

In this take-home: front end only (Next.js 14, Tailwind, local JSON and images, deployed on Vercel). Everything else is the target design.